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Problems and Postulates: Kant on Reason and Understanding

ALISON LAYWINE

THE PURPOSE OF THIS PAPER is to think anew Kant's conception of reason and understanding, the relation between these two faculties and the principles that govern them. I am chiefly interested in the contributions of reason and understanding to the advancement of knowledge. Hence the focus of my paper, so far as reason itself is concerned, is the theoretical rather than the practical employment of this faculty. On the other hand, I think it is useful and fair to say, on Kant's account, that theoretical reason is itself practical—at least to the extent that we regard scientific enquiry as some kind of activity, namely a *research* activity, involving practices such as proof and experimentation, undertaken for the sake of a certain end, namely, the acquisition of knowledge. One of Kant's concerns in the *Critique of Pure Reason* is to explain how this activity can be profitably pursued. To meet this concern, Kant must explain first of all how we set research goals for scientific enquiry. For without any such goals,

Susan Neiman's book, *The Unity of Reason* (Oxford: Oxford University Press, 1994), was the original stimulation for this paper. (I presented some of the ideas in this paper for the first time at a symposium on her book held by the North American Kant Society at the Central-Division Meetings of the American Philosophical Society in April 1995). I am more indebted to Susan Neiman for the idea that the task of reason is to give us direction and motivation in scientific enquiry. I shall signal other points of agreement with her in what follows. I had the opportunity in June 1995 to present a more evolved version of the paper in Martin Carrier's colloquium at the Ruprecht-Karls-Universitaet in Heidelberg, Germany. I am grateful to Martin Carrier and the participants of this colloquium for stimulating discussion, which helped me bring the paper into sharper focus. I am, likewise, most grateful indeed for the stimulating discussion I had with Milos Arsenijevic, Brigitte Falkenburg and Hans Friedrich Fulda, on various trips to Heidelberg. I must thank Stephen Menn for tips on ancient Greek geometry. Finally, I must thank Wolfgang Carl, Michael Friedman, Ulrich Majer and an anonymous referee of the *Journal of the History of Philosophy* for very helpful comments on a previous draft.

we would have no cause to enquire into anything.[1] Kant must also explain where we can find, and how we can use, the means to advance these goals. For again, without any such means, the activity itself would be vain. As I shall argue, the ultimate goals of scientific enquiry spring from reason. The means for achieving these goals and their implementation is the task of our understanding.

We can see why Kant divides the tasks of reason and understanding in this way, if we follow Kant's own example and reflect on the nature of geometrical demonstration. The difference between problems and postulates in geometry is important to Kant, precisely because he believed that it could help clarify the relation between our two higher faculties of cognition. Reason sets problems for us, just as the ancient geometers did: to that extent, it sets the goals of scientific enquiry. The understanding gives us the means to solve these problems (insofar as a solution is possible), because some of its principles have the character of geometrical postulates. Just as Euclid's postulates tell us that we may construct circles and line segments to the end of solving problems (as, for example, the problem of constructing an equilaterial triangle), so the Postulates of Empirical Thought tell us what the understanding is licensed to do to the end of solving the special problems set by reason. If we overlook the difference between problems and postulates, says Kant, we shall confuse the separate spheres of reason and understanding. In that case, the activity of investigating nature will cease, on our own account, to be practically viable.

One thing that should emerge from this paper is the great significance of ancient Greek geometry for Kant—not just for his philosophy of mathematics

[1] One way to express part of the idea is to say that Kant wants to address a question about the source of our motivation for engaging in scientific enquiry. I trust it is clear, however, that Kant's question is not concerned with motivational psychology or the problem of keeping given individuals or groups of individuals interested in, and excited about, doing science—the sort of problem science teachers face when they are concerned to keep their pupils from dropping out of physics out of boredom. Kant's question is of a different order altogether. He wants to understand what makes it reasonable as such to engage in scientific enquiry and, moreover, to continue with it—quite independently of the contingent factors that have stimulated the interest or excitement of particular individuals. To that extent, Kant's concern is related to that of Francis Bacon in Book One of the *New Organon*. Bacon was concerned, among other things, to argue against philosophical presuppositions about the status of human knowledge and the nature of human understanding that would have implied that it makes no practical sense to engage in any enquiry at all. In a sense, we can take Kant to be picking up from Bacon: assuming that Bacon mounted persuasive arguments to defeat these presuppositions, the question now is to show positively and in detail what practically speaking keeps scientific enquiry going and what makes the motor driving it intrinsically reasonable. It would not surprise me if Kant did actually see himself as picking up the *relais* from Bacon, if for no other reason than that Kant used a passage from Bacon's *Instauration magna* as a motto to the second edition of the first *Critique* and there are echoes of Bacon's work in the preface to the second edition (elsewhere too, I am sure, if one looked more closely than I have just yet).

properly speaking, but also for his conception of the way reason operates as a whole.[2] The evidence suggests that Kant used ancient Greek geometry as a clue or *Leitfaden* to unlock the workings of our higher cognitive faculties. To be sure, Kant says explicitly that Aristotelian logic plays the role of *Leitfaden*. But logic could do no more than indicate the *formal* constraints on our faculties. It could not help Kant see how reason and understanding cooperate to advance our knowledge of objects. Kant apparently believed that geometry could help him see this.[3]

My paper is divided into two sections. The first section is devoted to reason; the second is devoted to our understanding. In the last few pages, I shall sketch a picture of how the two faculties work together on Kant's account.

1.

Kant says that the distinctive feature of reason is its concern for what he sometimes calls "systematisation" or "systematic unity" (A645/B673; A694/B722–23). Reason takes a certain interest in the knowledge acquired by our understanding. This interest is expressed as an ideal to which such knowledge is expected to conform. As Kant himself puts it, reason seeks "a complete unity in the knowledge obtained by the understanding, by which this knowledge is to be not a mere contingent aggregate, but a system connected according to necessary laws" (A645/B673). Kant speaks here in the most abstract terms. But judging from the context and illustrations, his idea must be something like this. Our knowledge might very well increase, as we discover new laws of nature. But reason itself would not be satisfied with our discoveries until we could show that

[2] Of course, this raises a historical question, namely, what were Kant's sources on ancient geometry and what was the nature of these sources? One possible source is Lambert's philosophical writings. Lambert seems to have the same understanding of the Euclidean postulates as Kant does, and moreover he stresses the importance of having some kind of analogue of the postulates in metaphysics. See, for example, the opening chapter of the first volume of *Anlage zur Architektonic*. I suspect that Kant might have been stimulated by Lambert's ideas, but I cannot pursue my hunch here.

[3] To be sure, Kant denies geometrical methods can be adapted to metaphysics: hence geometrical construction in intuition plays no part in this science. See the *Preisschrift* of 1764 and the "Methodenlehre" of the first *Critique*. Nevertheless, Kant speaks unequivocally of "metaphysical constructions" in the preface to the *Metaphysical Foundations of Natural Science* (4.473.7–10). It is also interesting to note that the presentation of the *Critique of Practical Reason* is ostentatiously geometrical. We not only find theorems, but problems (of construction?) as well. Note the following "Aufgabe": "Given that the merely legislative form of maxims is alone the sufficient determining ground of a will, to find the character [*Beschaffenheit*] of that will which is determinable through it alone" (5.28.31–33). Likewise in §6: "Given that a will is free, to find the one law suited to determining it necessarily" (5.29.12–13). Geometrical construction may not be the proper way of solving *these* problems, but it is significant of Kant's regard for geometry, as reflecting the nature of demonstration as such and the operations of reason as a whole, that he sets problems, as it were *more geometrico*, in the *Critique of Practical Reason*.

these laws are all special cases of some higher, more fundamental principle. Indeed, it would not be completely satisfied until we had produced the highest principle of all—that principle from which all the laws of nature might be seen to follow. Systematic unity is a hierarchy of laws proceeding from a common ground. To say that reason takes an interest in systematic unity is to say that it urges us to find such a hierarchy in nature.

In order to clarify reason's contribution to scientific enquiry, we must determine the status of its interest in systematic unity. To that end, some preliminary remarks are in order.

The task of reason, according to Kant, is always to demand the conditions of our judgment. This is plain, he says, if we consider reason in its logical function, which is to treat any judgment of the understanding as the conclusion of a syllogism and to demand the conditions of this conclusion, namely, the major and minor premisses from which the conclusion is supposed to follow (A303–305/B359–61, A330–31/B386–87). Now reason admits no arbitrary constraints on its demands. So even after I have stated the conditions of my judgment in premisses of the appropriate syllogism, reason will again demand the conditions of my judgment. In other words, I shall have to state the major premiss of my first syllogism as the conclusion drawn from the premisses of a *prosyllogism;* then the major premiss of the prosyllogism will have to be stated as the conclusion of yet another prosyllogism, and so on and so forth. Reason will never be satisfied until I have produced the whole series of conditions, because a certain maxim seems to underlie all its logical operations, namely "to find for the conditioned knowledge obtained through the understanding the unconditioned whereby its unity is brought to completion" (A307/B364). Thus a consideration of the logical operations of reason is supposed to show us that reason is the faculty that seeks the unconditioned.

The interest in systematic unity comes back into the picture when we try to learn something from the preceding considerations about the *real* use of reason, namely, with respect to the *content* of our knowledge, rather than the logical order of our judgments in a series of syllogisms. Our concern, then, is the attitude and relation of reason to the objects of knowledge obtained by the understanding. Two considerations are relevant here.

First, Kant points out that the logical maxim of reason to seek the unconditioned cannot be applied, unless reason makes a certain assumption as a matter of principle, namely, that "if the conditioned is given, the whole series of conditions, subordinated to one another—a series which is therefore itself unconditioned—is likewise given" (A308/B364). Unless reason assumes that the whole series of conditions will become available to it, it cannot reasonably seek the unconditioned—or so it seems on the face of it. Second, the Transcendental Deduction is supposed to have established that the understanding can

have knowledge of objects only insofar as its pure concepts apply to objects of a possible experience. Then, by application of the assumed principle just mentioned in respect of the objects of our knowledge, reason may be seen to demand the whole series of conditions *in the realm of experience or the sensible world.* Now the conditions in the sensible world are laws of nature. So reason's demand for the whole series of conditions in the sensible world translates as a demand for systematic unity. This is because the unconditioned is presumed by reason to provide the ground from which all the laws of nature can be seen to follow.

Now reason's demand for systematic unity cannot be made without some kind of qualification. For the Antinomy is supposed to show that we can never meet a demand for the unconditioned. As a result, Kant says that reason must amend its principle—the principle, namely, that when the conditioned is given, the whole series of conditions is given forthwith.[4] Kant's remarks about the proposed amendation are instructive. He says, for example, that "[the principle of reason] preserves its validity, not indeed as the *axiom* [*nicht als* Axiom] that we think the totality as actually in the object, but as a *problem* for the understanding [*sondern als ein* Problem], and therefore for the subject, leading it to undertake and to carry on, in accordance with the completeness prescribed by the idea, the regress in the series of the conditions of any given conditioned" (A508/B536). The language here, I surmise, is that of geometry. So our question is how to understand the assumed distinction between geometrical axioms and problems. If we reflect on this distinction, we can bring into focus the nature and significance of geometrical problems for Kant. This will permit us to bring into focus the contrast between problems and postulates, which, in turn, will help us clarify the Kantian distinction between regulative and constitutive principles of reason.

On Aristotle's account, the axioms or common notions in Euclid's *Elements* are immediately self-evident truths, stated as first premisses in all of the demonstrative sciences (*Post. An.* 1.10.76b). Aristotle's favorite example is the axiom that equals taken from equals always leave equals. On the other hand, geometrical *problems* are to be understood, in the first instance, by contrast with geometrical *theorems.* A theorem states a universal truth about a whole class of figures. The Pythagorean Theorem tells us, for example, that, in any given right-angle triangle, a certain relation obtains between the square of the hypotenuse and the sum of the squares of the other two sides. By contrast, a

[4] In what follows, I shall refer to this assumed principle as "the principle of reason." I shall *not* use it to refer to Leibniz's so-called principle of *sufficient* reason, though there may well a relation between Leibniz's principle and the assumed principle of reason that is properly Kant's concern in the Transcendental Dialectic.

problem asks us *whether we can produce a given construction.* The first proposition in Book One of Euclid's *Elements* is stated as a problem, namely, to construct an equilateral triangle on a given finite straight line.

We know from the Discipline of Pure Reason that Kant takes the language of geometry seriously (B727–38). If, in the passage I cited from A508/B536, he is using this language in the traditional sense of the terms,[5] his point is that we may not take as an axiom or self-evident first premiss in every demonstration of pure reason the principle that, once the conditioned is given, the whole series of conditions, and hence the unconditioned itself, will be given forthwith. Accepting this principle as an axiom about objects of knowledge will throw reason into conflict with itself. So we must treat the principle as a problem set by reason itself for the understanding to solve. Thus the principle may be stated *problematically* as follows: *to see whether,* given the conditioned, one may not also find the condition of the conditioned and the condition of *that* condition, and so continue in the series of conditions, never stopping before you reach the unconditioned or complete systematic unity in the realm of experience.

Now the statement of a geometrical problem makes no presumption that we can succeed in constructing the assigned figure. In this respect, the statement of a problem stands in sharp contrast with the statement of a *postulate.*

Euclid's postulates declare that a certain procedure can be carried out.[6] Kant himself makes note of this in a passage from the second *Critique.* He says that "pure geometry has postulates as practical propositions that contain nothing, however, except the presupposition that one *can* do something, should it ever be required of us to do so" (5.31.1–3).[7] The third postulate in the first book of Euclid's *Elements* declares, for example, that a circle can be described with any center and distance. But as fondly as Thomas Hobbes

[5] Context suggests that Kant is using "axiom" at A508/B536 in the Aristotelian sense of a common notion. For the issue here is precisely whether to treat the principle of reason as an ultimate, self-evident first premiss in all of our demonstrations. As Michael Friedman has pointed out to me, Kant *seems* elsewhere to use "axiom" *not* in the sense of a *common notion,* but rather in the sense of a *postulate.* See, for example, B204–205, B760–61 and B16–17. It would be more grist for my mill, if it turned out that Kant is using "axiom" in the sense of "postulate" at A508/B536. But context suggests that he may well mean "common notion"—at least in this passage.

[6] In fact, this is obviously true only of the *first three* and the *fifth* postulates. I leave open the question whether the fourth is also constructive. I have so far not been able to determine what Kant makes of it.

[7] In the Antinomy section of the first *Critique,* Kant characterizes Euclid's second postulate in precisely the same terms (without, however, using the word "postulate"): "When we say, 'Draw a line', it sounds indeed more correct to add *in definitum* than *in infinitum.* Whereas the latter mean that you *must* not cease producing it—which is not what is intended—the former means only, produce it as far as you *please;* and if we are referring only to what it is in our *power* to do, this expression is quite correct, for we can always make the line longer, without end" (A511/B539).

might have hoped to solve the problem of squaring the circle (i.e., of finding a square equal in area to any given circle), even he did not presume that success in the venture was guaranteed from the outset. Likewise, reason offers no guarantee to the understanding that it can or will succeed in coming up with the goods. It *postulates* neither the unconditioned nor systematic unity.[8] Instead, it requires the understanding to continue working its way back through the series of conditions, forbidding it—as Kant explains—to bring the regress to a halt arbitrarily. No condition discovered so far may be accepted as the last in the series (A509/B537). Treated as a problem, the principle of reason is indeed, as Kant puts it, just "a rule prescribing a regress in the series of conditions of given appearances" (A509/B537). Moreover, if we consider again the contrast between a postulate and a problem, we can see that this rule is not *constitutive* of anything.

Precisely because Euclid's postulates declare that the basic procedures of geometrical construction can be carried out and iterated as many times as we like, they are, in some sense of the word, constitutive of the whole science of

[8]This is *not* to say that reason *never* postulates. When it has properly disciplined itself, it postulates what it has been expressly permitted to postulate. Since it has been expressly forbidden to postulate the unconditioned, it does not postulate the unconditioned. It is important to keep in mind the meaning of "postulate" in *geometry*. "Postulate" is our translation of the Greek "aitema." "Aitema" means that which you may rightfully demand be granted to you, for the purposes of demonstration. Since Euclid's third postulate tells us that we can construct a circle of any center and distance, we may rightfully demand that any objections about the possibility of such a construction be given up. Just to the extent that we make this demand, on the authority of Euclid's postuate, we are postulating circles. (I am grateful to Stephen Menn for helping me clarify the geometrical example.) On my interpretation of Kant's view, reason too may rightfully demand certain things, notably that the understanding conduct the synthesis of the manifold according to the principles laid out in the System of Principles. The System of Principles permits the use of the understanding according to these principles, just as Euclid's third postulate, say, permits the construction of circles. Thus reason may *demand* such use of the understanding—to the end which it prescribes, namely, pursuing the empirical regress of conditions so as to bring us closer and closer to complete systematic unity. This, I believe, explains passages such as this: "[The principle of reason] . . . serves as a rule, *postulating* [*postuliert*] what we ought to do in the regress, not anticipating what is present in the object as it is in itself, prior to all regress" (A509/B537). Reason does not postulate systematic unity as such, for it has no license to do so. Rather, it postulates or demands that the understanding do whatever is in its power to bring us closer and closer to systematic unity. It makes this demand on the basis of rights conferred on it by the Transcendental Analytic. Bear in mind, however, that Kant sometimes uses the word "postulate" equivocally. Sometimes Kant is very clear about this. Thus he states expressly in the *Critique of Practical Reason* that the so-called "Postulates of Practical Reason" are *not* postulates in Euclid's sense (5.11.28–41). Sometimes, however, Kant is not clear about possible equivocation—in which case, he is just being sloppy. The worst example is at B709. Kant has been saying that reason "gives an object" to its idea of systematic unity. This object, he says, "is postulated . . . *problematically*." Since a postulate presumes that we can succeed in carrying out a procedure, and since a problem makes no such presumption, it makes no sense to postulate something problematically. Nevertheless, Kant's point is reasonably clear from context: we must treat the assumption or demand for systematic unity in nature as a problem for ourselves.

geometry. For one thing, without the basic procedures of construction, no demonstration in this science would be possible. Euclid's demonstrations in the *Elements* are all based on figures constructed in accordance with the postulates. But for another thing, the postulates can be understood as licensing the activity whereby the objects of geometry are given to us in the first place—or so they are understood by Kant himself. Kant tells us in the *Critique of Pure Reason*, for example, that "in mathematics a postulate means the practical proposition which contains nothing save the synthesis through which we first give ourselves an object and generate its concept—for instance, with a given line, to describe a circle on a given plane from a given point" (A234/B287).[9]

[9]Here I note a puzzle in this passage. Kant says unambiguously that the objects of geometry and our concepts of these objects are first given to us by the postulates. Since Kant takes a definition to be just a presentation of "the complete, original concept of a thing within the limits of its concept" (A727/B755), geometrical definitions apparently depend on, and proceed from, the postulates. One consequence seems to be that, on Kant's view, the order of presentation at the beginning of Euclid's *Elements* is wrong. Euclid opens the *Elements* with the definitions. Then he gives us the postulates. Kant's view seems to be that the definitions of straight line, circle, etc. merely present us with the concepts we have acquired from the postulates. So it would also seem that the definitions should come after the postulates. And yet Kant makes it clear elsewhere that the postulates depend on the definitions and that the definitions somehow give us our concepts of mathematical objects. See the letter to Marcus Herz, 26 May 1789: "Furthermore, the possibility of a circle is not merely *problematic*, dependent, as it were, on the practical proposition 'to inscribe a circle by the movement of a straight line around a fixed point'; rather, the possibility is *given* in the definition of the circle, since the circle is actually constructed by means of the definition, that is, it is exhibited in intuition, not actually on paper (empirically) but in the imagination (*a priori*). For I may always draw a circle free hand on the board and put a point in it, and I can demonstrate all properties of the circle just as well on it, presupposing the (so-called) nominal definition, which is in fact a real definition, even if this circle is not at all like one drawn by rotating a straight line attached to a point. I assume that the points of the circumference are equidistant from the center point. The proposition 'to inscibe a circle' is a practical corollary of the definition (or so-called postulate), which could not be demanded at all if the possibility—yes, the very sort of possibility of the figure—were not already given in the definition" (*Philosophical Correspondence, 1759–1799*, trans. A. Zweig [Chicago: University of Chicago Press, 1986], 155). I am not sure how to solve this puzzle. Hans Friedrich Fulda has suggested to me that the definitions serve to assemble the marks of a given *definiendum* and are therefore the original presentation of a given concept. To that extent, they are first in the order of presentation. But, then, the relevant postulate is what first makes the concept concrete—concrete, insofar as it states explicitly that the *definiendum* can actually be produced in space. Thus Euclid's third postulate tells us that we can describe circles of any center and distance. We can understand the postulate to say that the concept presented to us in the corresponding definition is, after all, practically realizable in pure intuition, i.e., that we can give an instantiation of it in pure intuition. To that extent, the *postulate* first presents the concept to us, namely, as realizable in pure intuition. There is one thing that slightly complicates this solution to the puzzle. As Emily Carson points out, the definitions themselves are grounded in pure intuition. This means, among other things, that their realizability or instantiation in space is already guaranteed—without the postulates (which fact also helps explain why Kant characterizes postulates in the letter to Herz as "practical corollaries" of definitions). But, then, I believe that we may understand the postulates, as Fulda suggests, to make this possibility of instantiation in space *explicit*, and indeed explicit *for the first time*. After all, the possibility of the concrete instantiation of

Geometry is a science of objects, in the first instance (if not the last), precisely because of the postulates. Consider again the contrast between problems and postulates.

Like the postulates, some problems in geometry give us objects—figures we are assigned to construct—and help us form a determinate concept of these objects. This is true of problems that have a solution. A new object is given to us—and, with it, the corresponding concept—if we succeed, as Euclid does in Book Thirteen of the *Elements,* in solving the problem of constructing a dodecahedron inscribed in a sphere. But some problems do not admit of any solution. It is not the case that, given any circle, there exists a square of equal area. There *would* be such a square, if we could inscribe in the given circle a regular polygon whose sides were so small that they coincided with the circle's circumference. For we know that there exists a square equal in area to any given regular polygon. But here we encounter a difficulty. Procedures of construction, if carried out indefinitely, will allow us to inscribe regular polygons in the circle that *approach* the circle in area, without ever coming to equal it. Thus inscribe a square, for example, in the given circle. Then, on each side of the inscribed square as base, describe an isoceles triangle with its vertex on the arc of the smaller segment of the circle subtended by the side. The result is a regular polygon inscribed in the circle with twice as many sides as the inscribed square. Now repeat the procedure with the new polygon to construct yet another polygon with *four times* as many sides as the inscribed square. If we continue this procedure indefinitely, the small segments of the circle left over will be smaller than any assigned area.[10] But these procedures will not give us the desired regular polygon. Nor, then, can they give us the desired square. As a result, no determinate concept of either figure is possible

the definition of a circle is not explicit from the bare statement of the definition of circle. Precisely by making this explicit, Euclid's third postulate, for example, may be understood to generate the concept of circle—or, anyway, to present it to us for the first time as realizable in space. In any case, it is clear that there is some kind of chicken-egg relationship, on Kant's view, between postulates and definitions.

[10] Here I am following Heath's discussion of the "squaring of the circle" in Volume 1 of *A History of Greek Mathematics* (New York: Dover, 1981), 220–35. In fact, the procedure so described is attributed to Antiphon. Antiphon apparently thought that he would finally use up the area in the circle, whereupon he would have constructed a regular polygon inscribed in the circle whose sides would be so small as to coincide exactly with the circumference of the circle. Since we can construct a square equal in area to any given regular polygon, he believed he could then construct a square equal to the circle. Of course, you can't really use up the area of a circle in this way. As Heath notes, Euclid, who uses the same construction in Proposition Two, Book Twelve, describes more accurately what happens here. He says that, if we continue the construction far enough, the small sections left over in the circle will be smaller than any assigned area. The circle is thus the limit we approach as we continue the construction. See Heath, *History,* 222.

for us. The best we can do is to continue the procedure by exhaustion, constructing polygons that approach the circle as their limit.

While the postulates license activity that can always give us geometrical objects and that can always help us generate determinate concepts of these objects, not every problem can. For not every problem has a solution. It seems fair to say, then, that problems without a solution are not constitutive in Kant's sense. They are constitutive neither of objects nor concepts of objects. Rather, they apparently summon us to carry on towards some kind of assigned limit, like the area of the circle in the example we were just considering. In light of all this, we can see that the principle of reason should certainly not be treated as a postulate. On the other hand, it should not be treated as just any kind of problem either. It is best treated as a problem without any solution—like the problem of squaring the circle. For the principle of reason enjoins us to approach an assigned limit, namely, the unconditioned or complete systematic unity in nature. As Kant himself puts it, the principle of reason is "a principle of the greatest possible continuation and extension of experience, allowing no empirical limit to hold as absolute" (A509/B537).[11] However far along we proceed, no regular polygon I inscribe in the assigned circle will be equal to the circle in area. Likewise, no matter how much systematic unity we discover in nature, it will not be absolutely complete. Complete systematic unity is the limit we approach under the injunction of pure reason. It is, thus, a *regulative* principle (A509/B537). To that extent, it regulates our activity. Just as the problem of squaring the circle regulates our use of the method of exhaustion by directing us to the area of the circle, so the principle of reason regulates our investigative procedures by directing us to complete systematic unity in nature.[12]

[11] It should be emphasized that Kant is not saying that no systematic unity can be discovered in nature; he is saying rather that we must never be satisfied that as much systematic unity as we have discovered at any given moment is complete and absolute. Moreover, it is important to point out—as Hans Friedrich Fulda and Brigitte Falkenburg have emphasized to me—that, on Kant's view, it *is* possible to achieve complete systematic unity of *principles*. The project of setting up a table of categories, for instance, rests on the assumption that this is possible.

[12] A passage in the Appendix to the Transcendental Dialectic (A650–62/B678–90) seems to raise a problem. Kant says that "reason *presupposes* . . . systematic unity . . . on the ground that special natural laws fall under more general laws, and that parsimony in principles is not only an economical requirement of reason, but is one of nature's own laws" (A650/B678). Later, he says that "we have no option save to *presuppose* the systematic unity of nature as objectively valid and necessary" (A651/B679). It is not clear what Kant is saying. But he might be taken to say that reason must ultimately treat the demand for systematic unity as a postulate. To see what's going on, we must look very hard at what Kant means by "presupposes" in this context.

The "presupposition," "assumption," or "postulate" at issue seems to be an existential assumption to the effect that complete systematic unity obtains in nature (A650/B678). Kant apparently thinks that this existential assumption is somehow *practically significant*. If we did not make this assumption, it seems we could not engage in the activity of investigating nature. What, then, is the practical significance of the assumption? That is hard to say. But a couple of passages suggest that

Now I must note an interesting passage in Kant's discussion of the antinomy of pure reason, which seems to raise a difficulty for me. There is, indeed, much talk in the Transcendental Dialectic about problems. But the title of Section Four of the antinomy chapter is "The Absolute Necessity of a Solution of the Transcendental Problems of Pure Reason." Indeed, Kant claims that "no question which concerns an object given to pure reason can be insoluble for this same human reason" (A477/B505). It seems, then, that we can (and must) provide a solution not just for the problem of finding systematic unity, but for all of the so-called transcendental problems of pure reason. But in spite of appearances to the contrary, this passage tends, if anything, to support my reading.

Kant apparently thinks that we can solve the problems of pure reason if we understand how they arise in the first place. Since all of these problems have their origin in reason itself, the solution will always lie in understanding the nature of this faculty. Since this faculty hides nothing from our scrutiny, provided that we investigate it aright, we shall never despair of finding the solution. Consider the cosmological problems of pure reason—the questions raised by reason about the age, size, origin of the world and so forth. As Kant points out, all of these questions have to do with an object that has been given to us empirically, namely, the world, or rather the part of it known to us. Hence, the question or problem is to determine *a priori* how far the world extends in space, time, and the series of grounds and consequences. In Kant's own terms, reason wants to know *a priori* whether it can ever arrive at the absolute totality of conditions in the realm of appearances (A479/B507). Kant says that we can tell reason what it wants to know with the greatest

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the assumption expresses a *hope* that we can find systematic unity in nature—presumably as much systematic unity as is humanly possible (A650–51/B678–79). If a hope of finding some systematic unity is what's at issue, then my reconstruction of Kant's picture of reason is compatible with this passage. For such a hope is *not* a postulate in the technical sense of ancient geometry. A geometrical postulate tells us that we have the right to do something and that our expectation of success in doing this thing is *entirely justified.* By contrast, the *hope* of finding systematic unity, at issue from A650/B678 to A662/B690, *does not* assure success. How can it? The assumption that systematic unity obtains in nature is not an item of knowledge. We do *not* know; rather, we *assume* that systematic unity obtains in nature. We make this assumption, just because we hope to uncover as much systematic unity as we can. If we could not entertain *this* hope, we could not *reasonably* set the discovery of systematic unity as a problem for ourselves. If the hope and its underlying assumption were a postulate in the geometrical sense, it would tell us that we can be sure to find systematic unity if we go out looking for it, just as we can be sure, from Euclid's third postulate, that we can produce a circle if we use a compass in the appropriate way. (Note, however, that, though we must have *hope* of finding systematic unity, in order to be motivated to look for it, the hope all by itself does not move us to go out looking. I might hope to construct an equilateral triangle, but, unless I set this construction as a problem for myself, I will never do anything about it. Only the setting of the problem can actually move me. So Kant's discussion of hope does not preclude treating the demand of systematic unity as a problem. Indeed, it *requires* that we treat it thus.)

possible certainty (A479/B507). For the object of the cosmological ideas—unconditioned totality in appearances—can never be given as an object of possible experience. Kant thinks he has shown, from pure reason itself, that we can approach the unconditioned, but never reach it. So the solution to the cosmological problems set by reason consists in showing that no dogmatic solution is possible. "For so long as we obstinately persist in assuming that there is an actual object corresponding to the idea [the unconditioned given to us in the empirical synthesis]," says Kant, "the problem as thus viewed, allows of no solution" (A482/B510). Again Kant says that "the dogmatic solution is therefore not only uncertain, but impossible. The critical solution, which allows of complete certainty, does not consider the question objectively, but in relation to the foundation of the knowledge upon which the question is based" (A484/B512). The critical solution is precisely to show from the operations of reason itself that the dogmatic solution is impossible. This is just to say that reason's demand for the unconditioned—and thus its demand for absolute systematic unity as well—must be treated as a problem without a solution.

It is interesting that Kant looks again to mathematics in this passage to show us what kind of solution we can expect for the transcendental problems of pure reason. His example is Lambert's proof that pi is irrational. This proof is designed to address a certain problem, namely, whether we can give, in terms of rational numbers, the proportion between the diameter and the circumference of a given circle. Kant points out that this problem had never been solved before, and he asks whether any uncertainty remains about the prospects of a solution. The answer to this question is no. "Since no adequate solution in terms of rational numbers is possible, and no solution in terms of irrational numbers has yet been discovered," he says, "it was concluded that at least the impossibility of such a solution can be known with certainty, and of this impossibility Lambert has given the proof" (A480/B508).[13] Lambert succeeded in showing—with the greatest certainty—that the problem stated above admits no solution. Kant thinks of himself as having shown with just as

[13] It is not entirely clear to me what Kant thinks Lambert has shown. At a minimum, he is saying this: Lambert proved that we cannot give the proportion of the diameter and the circumference of any given circle, in terms of *rational* numbers. Lambert did, in fact, prove this. But Kant *might* be understood to say something else as well: Lambert proved that we cannot give this proportion in *irrational* numbers either. What might Kant mean by "irrational" numbers? Perhaps he means the "irrational radical quantities," which are the object of a conjecture Lambert makes at the end of his proof. These "radical quantities" are quantities expressed as roots: square roots, cube roots, etc. Thus Kant is saying that Lambert proved that the desired proportion cannot be given in terms expressed as roots. Note, however, that Lambert did not prove that this is the case. He stated it as a conjecture. See *A Source Book in Mathematics, 1200–1800*, ed. D. J. Struik (Princeton: Princeton University Press, 1986), 374.

much certainty that the problem in metaphysics of finding the unconditioned or complete systematic unity in nature has no greater chance of being solved than Lambert's problem in mathematics.

Let us pause to consider the significance of all this. The focus of Kant's attention is a human activity, namely, scientific investigation of nature. Kant's question is how this activity can be profitable for us. His answer, in the first place, is that the demand for systematic unity must be treated as a regulative rather than a constitutive principle, i.e., as a *problem without solution* rather than a *postulate.* Why? Under this condition, scientific enquiry will never lack for a goal, since it will be perpetually directed and moved towards absolute systematic unity as its limit.

Whereas a postulate *gives* us an object and helps us form the concept of this object, a problem *proposes* an object for us to look for. So too a problem without a solution. But the object proposed by Kant's problem without a solution is unlike any other, namely, the final end of all scientific enquiry. The idea of the unconditioned or complete systematic unity in nature points us in the direction of the very object that, as investigators of nature, we must want to find and know, whatever else we want to find and know. Particular research goals may be set for us by a given field of research, given the state of knowledge or ignorance prevailing in it at the moment. No doubt such goals should also be regarded as problems—problems that admit of solution to the extent that we actually manage to reach them.[14] But Kant presumably wants to say that these problems themselves attract the interest of reason only insofar as we are persuaded that their solution will advance our discovery of greater and greater systematic unity.

It could not have occurred to Newton to set as a goal for research the cause of the tides and the motions of the comets through all of space, if the physics of his time had not advanced far enough and had not suggested of itself that it would be plausible to study such phenomena. But that these phenomena should have excited Newton's interest and actually determined the course of his research—these and no other phenomena, which might have been studied just as plausibly given the state of physical theory in Newton's day—this could only be explained by the interest reason itself takes in systematic unity. For having established the law of universal gravitation as the cause of the motions of the planets in our solar system and the fall of heavy bodies at the earth's surface and so on, the most natural thing for Newton to do next, on Kant's conception of reason, was to investigate those phenomena most suggestive to him of even greater systematic unity in na-

[14] If we do not manage to meet them, it may very well be because they are of a special kind of problem without a solution, namely, dead ends.

ture, i.e., those phenomena likely to be further effects of the same cause. Thus Newton succeeded in showing that the tides and the motions of the comets are also the effects of gravitation. Insofar as we treat reason's demand for systematic unity as a problem without solution, it helps direct us to those problems whose solution in a given field of research would be most interesting from the standpoint of pure reason itself. When we have found a solution to *these* problems, it leads us on to the next ones. For it remains itself a task to complete, however much progress we have made in any given branch of scientific enquiry.

Precisely for this reason, scientific enquiry will always remain a going concern. The very fact that the demand for systematic unity must be treated as a problem without a solution ensures that the activity of investigating nature will never come to an arbitrary halt. This is highly paradoxical on the face of it. For one would think that, if we could establish the impossibility of solving a given problem, all our interest in working on the problem would vanish immediately. Who would be fool enough to try to give, in terms of rational numbers, the proportion between the diameter and the circumference of a given circle after Lambert had shown conclusively that this proportion cannot be expressed in such terms? If Kant can show as conclusively as he claims that we can never expect to find complete systematic unity in nature, why does he imagine that we can remain interested in it at all?

The answer has to be that not all problems without a solution are alike. Kant's problem surely distinguishes itself from Lambert's in the following respect. Kant's problem is some kind of ideal; Lambert's is not. The demand for absolute systematic unity is the standard against which we may judge the relative perfection of any knowledge we have acquired so far. The fact that human knowledge can approach perfection without fully achieving it does not make the standard any less compelling. For the standard permits us to discriminate between knowledge in a more perfect state and knowledge in a less perfect state; it teaches us always to prefer the former over the latter.[15] As Kant himself says, speaking of the ideals of pure reason: "Although we cannot concede to these ideals objective reality (existence), they are not therefore to be regarded as figments of the brain; they supply reason with a standard

[15] That said, there seems to be a difficulty in Kant's treatment of this ideal. We cannot set absolute systematic unity as an ideal goal, unless we can represent it to ourselves. We must know what we are ideally supposed to seek, otherwise there can be no *informed* appeal to the standard. But it is Kant's view that we cannot concretely represent this ideal to ourselves. We cannot know what we are ideally to seek—not in any robust sense of "know." Our ideal is in fact nothing more than a *call* for something, the something being specified only in the most formal terms. A mere call, however, is not yet an ideal—as Hans Friedrich Fulda has pointed out to me. Kant's treatment of the empirical employment of reason seems to be incomplete at best.

which is indispensable to it, providing it, as they do, with a concept of that which is entirely complete in its kind, and thereby enabling it to estimate and to measure the degree and the defects of the incomplete" (A569–70/B597–98). Precisely because the demand for systematic unity is an ideal of pure reason, it teaches us to be dissatisfied with the current state of our knowledge. Hence it always goads us to seek improvement. It thereby ensures that scientific enquiry remains a going concern. On the other hand, if we are stopped in our tracks by Lambert's proof that it is impossible to give, in terms of rational numbers, the proportion between the diameter and the circumference of a given circle, this is only because Lambert's proof does not give us an ideal of pure reason.

In short, Kant insists on treating reason's demand for systematic unity as a problem without a solution and not as a postulate, because otherwise scientific enquiry will lack motivation. One way to see the point is to consider the following question. Would scientific enquiry lose any of its motivational force as a human activity if we treated the demand for absolute systematic unity in some other way than Kant does in the passages we have been considering from the first *Critique*? What would happen to scientific enquiry if we treated this demand in the way of the *early* Kant?

The theme of systematic unity comes up in a number of Kant's early writings. It is very clearly articulated in *The Only Possible Basis for a Demonstration of the Existence of God* of 1763. "There are," says Kant, "many laws of nature whose unity is necessary, i.e., where the very same ground of agreement underlying one law also makes other laws necessary" (2.106.12–14). The idea is that some kind of unity arises when different laws proceed from the same principle. Kant says that such unity is necessary, because the relationship between these laws and the higher principle is that of ground and consequence and because, on Kant's account, this relationship is always a necessary one. By contrast, unity in nature is contingent if different laws proceed from different grounds (2.106.20–25). Kant gives us examples to make his idea more concrete. Living beings are supposed to be an example of contingent unity. For Kant says that we cannot show, for example, that the different senses in a human being are governed by different laws that proceed from one and the same higher principle. For all we know, the different organs of sense have been specially adapted to perform their function by separate acts of God's will. On the other hand, material nature is supposed to be an example of necessary unity. For material nature is governed by all sorts of laws that can be traced back to a fundamental principle. The motions of the heavens, the motions of the comets through all of space, the tides, even the shape of the earth are all to be explained ultimately as the effect of Newtonian gravitational forces and forces of repulsion at work in every particle of matter under the laws of motion (2.106.32–107.1).

Now the early Kant understood systematic unity, in the first instance, to be a task for God's intellect. Kant was a philosophical optimist. He presumed that God is infinitely wise and that God's wisdom commits God always to act for the sake of the best. Kant also thought that systematic unity is the highest perfection finite things can have. These assumptions led him to conclude that nature exhibits systematic unity, because God set himself as a task to create a world in accordance with his wisdom. Thus God decided to create matter first of all, because his intellect recognized in matter the greatest capacity to exhibit systematic unity and thereby to form a world capable of supporting rational life. To the extent that systematic unity is a task for *God's* intellect, it is also a task for *our* intellect. We seek to understand the world and to learn as far as possible the scope and content of God's wisdom. The question in light of all this is why systematic unity understood in these terms cannot rationally motivate and direct scientific enquiry. Here we have a goal for research, namely, to see the effects of God's wisdom in creation. Moreover, the goal is a reasonable one, since God's wisdom, being most perfect, represents an ideal standard for human reason. Having a reasonable goal, we shall not lack for motivation or direction.

But quite apart from any worry about antinomies concerning the age and size of the world and the existence or not of a necessary being, the picture of systematic unity painted by the early Kant may well introduce an element of arbitrariness into scientific enquiry.

From the standpoint of the later Kant, the early Kant treats demands for systematic unity as postulates. For the early Kant postulates that we can discover systematic unity in creation with sufficient time, patience, and cleverness. Such postulates may be justified by knowledge of God's nature, namely, that God exists, that he is supremely wise and that he gave the human mind everything it would need to uncover his laws at work in creation. Now, as we have seen, a geometrical postulate declares that a certain procedure can be carried out and that a certain object will thereby be given to us. But notice that that is all that a postulate ever declares. It will not tell us, for example, how many times to carry out the procedure. It leaves that up to us, because our decision to carry out the procedure or not, and if so, how many times in succession, will depend on what we want to do. *That* will be determined by the statement of the theorem we are trying to prove or the problem we are trying to solve. Now these considerations are relevant to demands for systematic unity treated as postulates. These postulates tell us, in effect, only that we *can* do something. Depending on how we disambiguate them, they tell us either that we *can* ultimately find the ground of *all* natural phenomena or (more modestly and more plausibly) that we *can* find the ground of *any given* natural phenomenon (while leaving it open whether all of them will or can be ex-

plained some day in the future). But even if such postulates could motivate us to find the ground of *some one* natural phenomenon, they could not tell us to carry out the search again for the ground of the next phenomenon or how many times to carry on the search thereafter. That would be left up to us. We might very well carry on, but there would be no reason not to call it quits. By treating demands for systematic unity as a postulate, not only does the early Kant unwittingly raise a question about the source of our motivation in scientific enquiry, he invites us to call the enquiry to an arbitrary halt.[16] If we follow the later Kant and treat the demand for systematic unity as a problem, and indeed as a problem without a solution, reason enjoins us not to call an arbitrary halt in the regress of conditions. For we may not be satisfied with anything less than the unconditioned itself. From the standpoint of the later Kant, then, the early Kant may well introduce an element of practical arbitrariness into our picture of scientific research.[17]

Textual evidence from the *Critique of Practical Reason* tends to support my conjecture about the later Kant's reading of his earlier self. I have already quoted from the passage I have in mind. But it is worth thinking about the passage as a whole.

In Part One, Book One, Section One, §7 of the second *Critique,* Kant states the fundamental law of pure practical reason: act in such a way that the maxim of your will can always also hold as principle of a universal legislation. This is a *practical* principle; moreover, it is a practical principle issued by *pure* reason. But it is also supposed to be unique. Thus Kant makes a special point of distinguishing it from other pure practical principles, namely, the postulates of Euclid. The key thing is that the fundamental law of pure practical reason is supposed to command us unconditionally. As Kant explains: "But here [in the case of the fundamental law] the rule states that one must proceed absolutely in a certain way. The practical rule is thus unconditioned [*unbedingt*], therewith as categorically practical" (5.31.6–7). Euclid's postulates are different: "Pure geometry has postulates as practical propositions that

[16] As Martin Carrier points out, one might well think that we would be more effectively motivated to seek the unconditioned if we could postulate both that this ground exists and that we can find it, given enough time, patience, cleverness, etc. Indeed, it may well seem that, treated as a postulate, the demand for systematic unity or the unconditioned has greater motivational force than if treated as a problem without a solution. But Kant thinks that nothing could be farther from the truth. Treated as a postulate, the demand brings scientific enquiry to an end, because it simply cannot motivate. The relevant authorities might license me to drive a car, as Euclid's third postulate permits me to construct a circle. But unless I have some independent reason to do so, I will never drive my car—much less construct a circle.

[17] Thus I find myself in agreement with Susan Neiman, who argues in *The Unity of Reason* that, from Kant's standpoint in the *Critique of Pure Reason,* rationalists such as Leibniz cannot explain what keeps us interested in scientific enquiry.

contain nothing other than the presupposition that we can do something if we should be required to do so. . . . *They are thus practical rules under a problematic condition of the will*" (5.31.2–6).[18] Unlike the fundamental law of pure practical reason, Euclid's postulates have no claim on us, unless we happen to have independent grounds for acting on them. These grounds are given, as Kant says so suggestively, under a "problematic condition of the will." The implication is that, as a condition for acting on any of Euclid's postulates, some problem has to be set for us—either a problem properly speaking, such as the problem of constructing an equilateral triangle on a given line segment, or a theorem *stated* problematically: to see whether we can prove, for example, that, in any given right-angle triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

2.

Now let's turn to Kant's conception of pure understanding. If we must treat the demand of reason for systematic unity as some kind of problem, it is tempting to treat the principles of pure understanding—some, if not all of them—as postulates. I propose, then, to examine Kant's claim that the postulates of empirical thought may be likened to the postulates of Euclid.

The postulates of empirical thought guide the use of our modal concepts—the possible, the actual, and the necessary. These concepts are distinctive, says Kant, because "they do not in the least enlarge the concept to which they are attached as predicates" (A219/B266). "Even when the concept of a thing is complete," Kant explains, "I can still imagine whether this object is merely possible or is also actual, or if actual, whether it is not also necessary. No additional determinations are thereby thought in the object itself . . ." (A219/B266). But if the pure modal concepts "do not in the least enlarge" any other concept of which they might be predicated, what function do they serve? Kant replies that they somehow indicate the origin of our *other* concepts in our different faculties of knowledge: "[The concepts of modality] only express the relation of the concept [of which they are predicated] to the faculty of knowledge [from which this concept proceeds]" (A219/B266). Elsewhere, Kant says that, though our pure modal concepts are not *objectively* synthetic, for the reasons indicated above, they are nonetheless *subjectively* synthetic, in that "they add to the concept of a thing, . . . of which otherwise they say nothing, the cognitive faculty from which it springs and in which it has its seat" (A233/B286). Thus we conceive of an object as *possible*, Kant tells us, if we represent it under the "formal conditions of experience," whatever they may be, laid down by pure understanding. Kant tells us that we conceive of an object as *actual*, if we repre-

[18] The emphasis is mine.

sent the object by means of perception and the laws of sensibility, whatever they may be. Finally, Kant tells us that we conceive of an actual object as necessary if we represent this object, given to us in perception, as causally related to other objects under causal principles of pure understanding (A218/B265–66).

Whatever precisely Kant thinks about the role of perception, sensibility, and understanding in all this is not yet clear. But the important point should be: namely, that the function of our modal concepts is to indicate the origin of our knowledge of an object in the appropriate cognitive faculty. Kant himself puts it helpfully and suggestively as follows: "The principles of modality thus predicate of a concept nothing but the action of the faculty of knowledge through which it is generated" (A234/B287). This statement is significant, because it ultimately explains why the modal principles should be likened to Euclid's postulates.

At this point in the discussion, Kant clarifies the meaning of the word "postulate." As I noted earlier, he tells us that "in mathematics a postulate means the practical proposition which contains nothing save the synthesis through which we first give ourselves an object and generate its concept . . ." (A234/B287). Euclid's third postulate does not add anything to our concept of the circle, as expressed in the definition of this figure. It tells us only that the construction of the circle can be carried out. For it "contains" or indicates the very activity by which a circle of any center and diameter can be constructed. Likewise, the modal concepts "do not in the least enlarge" the concept of a thing; rather they indicate "nothing but the action of the faculty, through which [the concept of the thing] is generated," as Kant himself puts it so helpfully.[19]

Kant's discussion of the modal principles is very interesting. But it is also so abstract that one still has to wonder why we should think of the modal principles as postulates in anything like Euclid's sense. It might help if we could understand more concretely what the modal principles are good for. Euclid's

[19] Notice, then, how important it is for Kant to specify in the preface to the *Critique of Practical Reason* that the postulates of pure practical reason are *not* postulates in the strict, mathematical sense: "But the expression '*postulate* of pure practical reason' might still cause an even greater misunderstanding if one should confuse the meaning of this expression with that of the postulates of pure mathematics, which postulates are characterized by apodeictic certainty. But the latter postulate the *possibility of an action* whose object one has already recognized *a priori*, theoretically with full certainty to be *possible*. The former, however, postulate the possibility of an *object* itself (God and the immortality of the soul) from apodeictic, *practical* laws, and thus solely to the profit of a practical reason. For this certainty of the postulated possibility is not at all theoretical; nor, as a result, is it apodeictic. In other words, it is not a necessity recognized with respect to an object; rather it is a necessary assumption with respect to the subject in the observance of his objective, but practical laws, and thus it is merely a necessary hypothesis. I could not find a better expression for this subjective, yet true and unconditioned necessity of reason" (5.11.28–41).

postulates permit us to construct lines and circles. What can we "construct" with Kant's postulates?

One way to answer this question is to consider how Kant applies the modal principles to the empirical concept of matter. For we may hope, as Kant himself says in the preface to the *Metaphysical Foundations of Natural Science,* to gain from such an application "examples (cases *in concreto*) in which to realize [pure concepts of the understanding] . . . , i.e., to give to a mere form of thought sense and meaning" (4.478.17–20). If we can give "sense and meaning" to the principles of modality, we might be able to understand more concretely why they should be taken for postulates and what they license us to do under that title. Now Kant already announces in the preface that his treatment of the modal principles in Chapter Four of the *Metaphysical Foundations* will allow us to determine the motion of a body or a particle "as an appearance of the external senses." While Euclid's postulates permit us to construct lines and circles, Kant's postulates will apparently permit us to construct frames of reference in which we may represent motion as either true or apparent.

The role of the postulates in discriminating between true and apparent motion has been stressed recently by Michael Friedman. As Friedman argues, Kant is consciously distinguishing himself from Newton in the following respect. For Newton, discriminating between true and apparent motions is a matter of discovery. We begin by assuming the existence of absolute space, time, and motion; we formulate the laws of motion relative to this framework. Then we set out to find the true motions—whether the sun revolves around the earth or the earth around the sun—by applying the laws of motion to the observable, apparent, motions described by Kepler's "phenomena." Since Kant himself denies absolute space and time, discriminating between true and apparent motion cannot be a matter of discovery; rather, it is a matter of construction—by some kind of procedure carried out *a priori* by the understanding.[20]

In the observation to the opening Clarification of Chapter Four in the *Metaphysical Foundations of Natural Science,* Kant puts it, in effect, as follows. The movable is given to us in appearances, i.e., in the manifold of empirical intuition, or not at all. But appearances all by themselves do not tell us whether the motion of something is true or merely apparent. Hence, our understanding must organize appearances so that we will have frames of reference in which true and apparent motions can be meaningfully distinguished from one another. Whatever the understanding must do to this effect is specified by the modal principles applied to the empirical concept of matter or the movable in space. As it turns out, these principles specify that the

[20] Friedman, *Kant and the Exact Sciences,* 136–65.

understanding should employ the analogies of experience, interpreted by the doctrine of body as the laws of motion. To the extent that we represent a body or particle as governed by the laws of motion, we may represent its motion as true rather than apparent. As Friedman argues, Kant apparently takes the laws of motion to be *constitutive,* rather than descriptive, of the spatio-temporal framework of Newtonian celestial mechanics.

How does this story about the postulates cash out? Consider the first modal principle as applied to the empirical concept of matter. This principle governs possibility and is stated as follows: "The rectilinear motion of a matter with respect to an empirical space, by contrast with the opposite motion of the space, is a merely *possible* predicate. The very same in no relation to any external matter, i.e., thought of *as absolute motion,* is *impossible*" (4.555.15–19). The stress here is on rectilinear motion, whether uniform or accelerated. Such motion is represented as relative. Suppose we have two bodies, the earth, say, and a rocket. Let's imagine, first, that the earth can be taken to be at rest with respect to the fixed stars and, second, that the rocket can be taken to accelerate in a straight line away from the earth. The first modal principle tells us that, under such conditions, there is no reason not to regard the earth itself, together with the fixed stars, as accelerating away from the rocket, taken to be at rest. It is simply a matter of perspective and convenience.[21]

Precisely insofar as we represent rectilinar motion as relative, the first postulate permits us to make an arbitrary choice. We may regard the rectilinear motion of a body, with the surrounding space and neighboring bodies at rest, as *possibly* true. We will actually go ahead and make this assumption, provided, of course, that we have some independent reason to do so. Thus the first modal principle is indeed a postulate in the relevant sense. If we tackle a problem, whose solution depends on making an arbitrary assumption about motion and rest, the assumption is licensed by the first modal principle, and thus we may carry on to solve the problem. Suppose the problem is to find the law of falling bodies, given nothing more than the phenomenon as observed by Galileo himself. Thus we observe that bodies fall with accelerated motion to the ground. We want to find a way to express this acceleration, putting aside for now, just as Galileo himself did, the question concerning the forces that might conceivably cause it. Our solution to the problem depends on an arbitrary assumption that the motion of a body falling to the earth, taken to be at rest, is *possibly* true. We could as easily regard the earth as rising towards a rock, taken to be at rest, suspended in mid-air. But if we find it more convenient to think of it the other way around, we do so—on the authority of the

[21] It would not be a matter of perspective, however, if the earth and the rocket were exercising mutual and opposite forces on each other. See below.

first postulate of empirical thought. Having made this assumption, we may solve the problem, so defined, provided that we subsequently think it through properly.[22]

Now consider the second modal principle applied to the empirical concept of matter. This principle governs actuality: "The circular motion of a matter, by contrast with the opposite motion of space, is a *real predicate* of the same; on the contrary, the opposite motion of a relative space, instead of the motion of the given body, is no real motion, but rather, when so taken, a mere illusion" (4.556.30–557.3). This modal principle gives us the right to regard motion ascribed to a body not just as *possibly* true, but as *actually* true. It gives us this right under the condition that our understanding applies the Second Analogy (interpreted as Newton's first two laws of motion) in cases of observed circular motion.[23] Under Newton's laws, we may represent a body as moving in a

[22] The story becomes more complicated as our understanding of the phenomenon becomes richer. If we have worked through *Principia*, Book Three, we know that the earth and heavy bodies are exercising mutual and opposite gravitational forces on each other. Hence, the earth is indeed rising (however little) to terrestrial bodies. The further story about mutual and opposite gravitational forces yields, in fact, a new, privileged frame of reference, defined by the center of mass of the earth/falling-body system, fixed pretty close to the earth's center. Given the new frame of reference, we must correct our initial assumption that the earth is at rest.

What this example shows, as Michael Friedman has emphasized to me, is that there is an important difference between Euclid's postulates and Kant's postulates of empirical thought. The former license constructive procedures that are carried out entirely *a priori*. Kant's postulates give license *a priori* for constructive procedures that take the empirically observable—the motions of, say, falling bodies as observed by Galileo—as their point of departure. We may, as he has suggested to me, look at it this way: pure understanding (in connection with pure sensibility) gives the rules of construction, perception gives the point of departure, and experience results when we apply the rules to our starting point in perception. On the other hand, the example also shows that Kant's postulates have something else in common with Euclid's postulates, which I have not yet properly stressed. Whether and how we apply one of Euclid's postulates depends not only on the problem we wish to solve, but also on *what is given* to us to start out with. I have a reason to act on Euclid's third postulate and to construct circles, if I wish to solve the problem of constructing an equilateral triangle on a given line segment. But the radius of the circles I construct will be determined by what's given to me, namely, the given line segment. Similarly, I have reason to act on Kant's postulates, if I wish to solve physical problems concerning true and apparent motion. But how these postulates are applied will, again, depend on what is given to me: in the first place, on what has been observed; in the second place, on what might or might not have already been established by the relevant physical theory. Moreover, subsequent application of the postulates may alter the way we think about what is given, so that we may be required to refine or correct our description of the systems that interest us.

[23] It is indeed odd that Kant should privilege circular motion here. Given that the second modal principle applied to the empirical concept of matter interprets the Second Analogy as Newton's first two laws of motion, one would expect that rectilinear accelerated motion should not be left out of account. We expect Kant to say that whenever we conceive forces to be in effect, the predicate of motion is real. Yet there is no mention of rectilinear motion at all in the statement of the principle. One might argue from the principle of charity that Kant must have intended to mention it, but somehow neglected to do so. But at least one thing telling against that move is that Kant explicitly mentioned "rectilinear" motion in the statement of the previous modal principle,

circle, just in case we also represent the body as suffering the effect of some kind of moving force. By the law of inertia, we have to represent the body as striving at every moment to move with uniform speed along a straight line tangent to its circular orbit. Since we represent the body as being constantly deflected from the path of its inertial motion, we must suppose, by Newton's second law of motion, that some kind of force is the cause of this acceleration. But by the same token, we represent empirical space surrounding the revolving body as unaffected by this force. Thus we may represent empirical space as actually at rest, and the body itself as actually moving in a circle. Insofar as we represent the earth as actually revolving around the sun, under the influence of some force, we may represent the fixed stars as actually at rest. We do not imagine that the fixed stars have an oppositely directed revolution around the earth. For then, to account for the same phenomenon, we would have to imagine that they undergo a horrendously great acceleration.

The second modal principle is also a postulate, for it too gives us license to take certain steps to solve problems. Consider the problem of finding the relation between the acceleration of the moon in its orbit around the earth and the acceleration of bodies falling at the earth's surface. Suppose that nothing more is given to us than the phenomena as observed by Newton. Suppose that, like Newton himself at Proposition 4, Book Three of the *Principia*, we know already that the moon accelerates inversely as the square of the distance towards the earth's center, without having shown that the earth in turn undergoes an equal and opposite acceleration towards the moon. The solution to our problem, with the given so defined, depends on regarding both the moon and the bodies falling at the earth's surface as *actually* falling towards the earth's center, taken to be *actually* at rest. If we now elect to regard the earth itself as rising towards terrestrial bodies imagined to be at rest in mid-air, we could not compare the accelerations of interest to us.[24] The second modal principle tells us that we may make this comparison, and how: we apply Newton's first two laws of motion to the moon. Thus we represent the moon's

apparently including accelerated rectilinear motion as well as inertial rectilinear motion. This seems to suggest that he intended some peculiar contrast between rectilinear motion, as treated by the first modal principle, and circular motion, as treated by the second modal principle. This is admittedly strange.

[24] If we are already in possession of the theory of gravitation, we know that the earth and heavy terrestrial bodies are mutually attracting each other, and hence falling towards each other. The earth is indeed rising, however little, to a body falling at its surface. Moreover, it is rising towards the moon as well. But at Proposition 4, Book Three of the *Principia*, Newton has not yet established that either the earth and the moon, or the earth and terrestrial bodies, are undergoing equal and opposite accelerations towards each other. Once he does, however, the earth/moon/falling body frame of reference will have to be redefined—in terms of the center of mass of the whole system, taken to be necessarily at rest somewhere pretty close to the center of the earth.

nearly circular orbit as composed of the moon's inertial motion and its fall towards the earth's center under the influence of a centripetal force varying inversely with the square of the distance. We cannot reasonably regard the moon to be at rest with the space surrounding it revolving in the opposite direction. To that extent, we may regard the moon as actually revolving around the earth. Hence, we may regard the center of the earth as *truly* or *actually* at rest in this frame of reference. As a result, we are no longer free to imagine the earth itself as rising towards terrestrial bodies hanging, as it were, in mid-air. Insofar as the second modal principle tells us what we may do to the end of solving this problem, it is indeed a postulate. It does not solve the problem all by itself. But that does not make it any less a postulate.

Finally, the third modal principle gives us the condition under which we may represent motions as *necessarily* true, namely, when the understanding applies Newton's third law of motion to appearances: "in every motion of a body, with respect to another body, an equal and opposite motion of this other body is necessary" (4.558.8–10).

Suppose we have two bodies, A and B. Suppose that B is apparently at rest with respect to all neighboring bodies and that A is apparently accelerating towards B at a certain rate. After a certain time, the bodies meet and resist mutual penetration. Kant can be understood to say that it would be highly arbitary, insofar as a collision takes place, to assign the whole acceleration, prior to the collision, to A and to say that B remained at rest. We must say neutrally that the distance between A and B was covered in a certain time at a certain rate. Then we must equally divide the whole acceleration between the two bodies, in indirect proportion to their mass. This is, in effect, just to apply Newton's third law to the observed phenomenon: $m_A a_A = - m_B a_B$. Even though B may be taken to be at rest with respect to all neighboring bodies, B has a true acceleration towards A. Moreover, this acceleration is necessary. To the extent that the one body is represented as participating in the whole acceleration, so must the other. On the other hand, to the extent that B may be represented, before the collision, as being at rest with respect to all neighboring bodies, these bodies too must be assigned the appropriate acceleration. Likewise, after the collision, except that, unlike B, we must represent the neighboring bodies as unaffected by A's impact. Then again, the common center of mass of the A-B system must be represented as being at rest.

Like the second modal principle, the third modal principle is useful for excluding rival assignments of motion and rest. But it also allows us to refine or correct assignments of motion and rest secured by the second modal principle. Thus, by application of the second modal principle, we might reasonably say with Newton at Proposition 3, Book Three of the *Principia* that the moon undergoes a real, inverse-square acceleration towards the earth and that the

fixed stars are really at rest. But by applying equality of action and reaction under the third modal principle, we are called upon to correct the assumption that the earth itself is at rest. Thus we learn that the earth in turn undergoes a real (however slight) inverse-square acceleration towards the moon. This shows us that the more precise assignment of rest must go to the center of mass of the whole earth-moon system (located pretty close to the center of the earth itself). As Friedman argues, the third modal principle gives us a privileged frame of reference—a center of mass frame in which the apparent motions turn out *necessarily* to be true motions.[25] Just to that extent, it too is a postulate in the relevant sense. If a problem we wish to solve depends on specifying a center of mass frame, we may call on the third principle and apply Newton's third law. In 1754 Kant had set himself the problem of figuring out whether the earth's axial rotation is slowing down. The solution to that problem depended on assigning equal and opposite inverse-square accelerations to the earth (and its oceans) and the moon. Otherwise, it would have been impossible to assign to the tides a necessary drag on the earth's rotation, much less to calculate the effects of this drag (1.183–193). From the perspective of the later Kant, the early Kant could not have solved his problem without applying the third modal principle of the *Metaphysical Foundation*—treated as a postulate in Euclid's sense.

Considering the modal principles, as applied to the empirical concept of matter, helps us understand more concretely why Kant holds that these principles are postulates. Whereas Euclid's postulates permit us to construct lines and circles, Kant's postulates permit us to construct frames of reference in which true and apparent motions can be distinguished from one another. Again, we can understand more concretely what Kant has in mind when he says that the modal postulates do nothing more than indicate the activity of the understanding by which a certain object is first given to us and its corresponding concept is first generated. The predicates "possible," "actual," and "necessary" add nothing at all to our concept of motion. Instead, they indicate, and indeed license, the activity of the understanding whereby true and apparent motions first become an object of experience for us. The activity in question is not the use of straightedge and compass, as one might expect in the case of Euclid's postulates, but rather the application of Newton's laws of motion. To the extent that this activity gives us true and apparent motions as objects of

[25] As Friedman points out, though, there can be no ultimate, absolutely privileged center of mass frame, for the simple reason that, on Kant's view, we can only approximate more and more precisely the center of mass of the whole universe, without ever establishing it once and for all. Thus every center of mass frame we specify can, and must, be corrected by subsequent applications of the third modal principle. This, I believe, may have paradoxical consequences for the modal principles themselves. But this is not the place to take up that discussion.

experience, we can do what would otherwise be impossible, namely, entertain a meaningful concept of such motions.

Now we might expect Kant's account of the postulates of empirical thought to have implications for the other principles of the understanding. The other principles of the understanding are not themselves postulates. Kant is explicit about that. Still, some, if not all, of them are surely that which the postulates are supposed to indicate and license, namely, different "actions" of the understanding through which concepts of objects are generated in the first place (A234/B287). This has to be true for the analogies of experience. As we have just seen, the postulates of empirical thought, as treated in the *Metaphysical Foundations,* indicate what the understanding can, and must, do if we are to have any meaningful concept of true and apparent motion: it must apply the laws of motion in the appropriate way. But the laws of motion themselves are just the analogies of experience applied to the empirical concept of matter. So though the analogies of experience are not postulates in their own right, they are the referent of Kant's postulates, namely, the action of the understanding indicated by the postulates for generating certain concepts.

What of the so-called axioms of intuition and the anticipations of perception? It seems fair to say that these principles are essential to the understanding whenever it puts the analogies to work. Consider again the application of the categories to the empirical concept of matter. The analogies, thereby interpreted as the laws of motion, presuppose the concept of force as the cause of acceleration. But forces can be stronger or weaker. In other words, they have "intensive magnitude." Thus any use of the laws of motion presupposes the necessary condition of all thought about intensive magnitude, namely, the anticipations of perception. Likewise, any use of the laws of motion presupposes the axioms of intuition insofar as we must represent moving bodies as having "extensive quantity": e.g., quantity of matter. To the extent, then, that the postulates of empirical thought indicate and license the understanding's use of the analogies of experience, they surely indicate and license its use of the other principles as well.

It follows that the analogies of experience, anticipations of perception and axioms of intuition stand to the postulates of empirical thought in something like the relation that the activity of describing circles and producing line segments stands to the postulates of Euclid. Describing circles and producing line segments are just the procedures indicated and licensed by Euclid's postulates. Euclid's postulates tell us that these are the procedures we can, and must, employ in order to solve problems in geometry. Likewise, the postulates of empirical thought tell us that the principles of the understanding listed above are the procedures we can, and must, employ in order to solve problems in natural philosophy, which depend on distinguishing true and apparent

motion. Precisely because the understanding is itself the source of both the postulates of empirical thought and the constructive procedures which they license, we may reasonably characterize it as a faculty of postulates—just as we characterized reason itself as a faculty for setting problems.

Now I shall try to determine Kant's philosophical motivations for conceiving of the understanding in this way. I have two suggestions to make.

Recall the difficulty Kant considers in the letter to Marcus Herz of 21 February 1772. Kant notes that he had characterized the intellect in his *Inaugural Dissertation* of 1770 in purely negative terms, i.e., as a faculty that represents objects, without ever being affected by them. As Kant admits, this raises a difficult question. How can pure intellect form any representations of objects? If it were affected by objects, it would stand in *some kind* of relation to them, and we might say that it forms representations of them precisely by virtue of this relation. But Kant denies himself this way out, because he insists that sensibility, and sensibility alone, is a faculty to be affected by objects. Though he might explain how *sensibility* can form representations of objects, he still has a problem with pure intellect. If the problem is, in the first place, to establish some kind of relation between pure intellect and the objects of its representations, Kant might well suppose that these objects are *created* by pure intellect. But Kant denies himself this way out too, because he says that God alone has the power to create (10.125). Unless he can explain how pure intellect can have any relation to objects at all, it would seem that the intellect can have no representations, nor any knowlege of objects.

In order to solve the dilemma in the Herz letter, Kant must find a third way of establishing the sought after relation, which requires neither that the intellect be affected by its objects, nor that it create its objects. On reflection, Euclid's postulates seem to give us a model of what we're after. Thus one can imagine Kant thinking to himself in 1772 and thereafter in the following terms. Is it possible that the understanding engages in activities something like those carried out by Euclidean geometers when they use the postulates in their constructions? If so, we may regard the understanding as making its objects possible in the first place, though not by any, as it were, divine act of creation. Thus it may become intelligible to us how *a priori* representations of objects are possible: such representations are themselves possible by the very same activity of the understanding that makes their objects possible in the first place. In just the same way, the activity of producing a line segment, as indicated in Euclid's first postulate, is the condition of possibility underlying any line segment and any thought of a line segment.

Of course, the problem in the Herz letter is not merely to explain how representations of the intellect relate *a priori* to objects; the problem is also to explain how, by such a relation, the intellect secures some kind of "*necessary*

agreement" between its representations and their objects (10.126). By virtue of such an agreement, our intellectual representations would never fail to specify how their objects are possible and (I infer) what their nature consists in. Again, Kant would likely have found it instructive to consider the example of Euclid's postulates. For the concept of line segment given by Euclid's first postulate cannot fail to inform us about the nature of its object. Knowing what a line segment is and knowing how to construct it are one and the same thing. That there is necessary agreement between such knowledge and its object is seen from the fact that anyone who knows what a line segment is cannot fail to construct a line segment at will. The necessary agreement is itself established by the fact that the knowledge and its object ultimately have the same source—the constructive activity licensed by the relevant postulate. If Kant can show that the intellect engages in activities analogous to this, he can show that the intellect is likewise justified in assuming that a necessary agreement obtains between its representations and their objects, and thus that knowledge of these objects is possible.

The significance of Euclid's postulates for the problem of justification in this context is all the more plain if we consider the following question. Why might one think that it is important to have postulates as well as axioms in the elements of geometry? Let's put the question in the following terms. What do we have to lose, on Kant's account, if we reformulated the postulates as existential statements about eternal objects? Thus Euclid's first postulate would say that there are line segments extending from any point to any point, and his third postulate would say that there are circles of any center and distance. What do we have to lose from such a reformulation? The answer, I take it, is any hope of addressing Herz-style difficulties, at least with respect to mathematics.

If we start out in the elements of geometry with a claim about the existence of certain eternal objects, we must then consider how these objects are given to us and how we form concepts of them. If we say that these objects affect the intellect somehow, we threaten the status of mathematics as a body of *a priori* truths. Again, sensibility alone, on Kant's account, is a faculty to be affected by objects. Thus any concept formed by an object affecting the mind is empirical. We cannot plausibly say that the human mind *creates* the eternal objects of geometry. As a result, we find the origin of our thought about these objects more and more mysterious. Shall we say that God benevolently planted concepts of circles and line segments in our mind and that these concepts happen to square with their objects—again, by an act of divine benevolence? If so, we bring down on our heads all the scorn Kant heaps on Crusius:

> Crusius believed in certain implanted rules for the purpose of forming judgments and ready-made concepts that God implanted in the human soul just as they had to be in

order to harmonize with things. . . . But the *deus ex machina* is the greatest absurdity one could hit upon in the determination of the origin and validity of our knowledge. It has . . . also this additional disadvantage: it encourages all sorts of wild notions and every pious and speculative brainstorm. (10.126; Zweig translation, 72–73)

The lesson to be drawn from these remarks is precisely that, once we start talking about eternal objects, we shall find it impossible to answer questions about justification without lapsing into some kind of "speculative brainstorm." By reconverting our axioms about the existence of eternal objects back into Euclid's postulates, justification does not pose any such problem for us.[26]

These considerations suggest that Kant may very well have come to think of the understanding as a faculty of postulates in order to solve the difficult problem, first raised in the letter to Herz, of explaining how our pure concepts relate *a priori* to objects and how they may be entitled to the claim of conveying knowledge of any such objects. But if my suggestion has any plausibility, it might also point out the direction Kant's work was to take from 1772 until the publication of the first *Critique*. For we may presume that the problem Kant then had to face was how to determine which activities of the understanding were such as to give us postulates in the relevant sense. Just as the ancient geometers debated amongst themselves about which constructions could be admitted into the elements of *their* science, so Kant has to figure out which activities of the understanding are appropriately constructive and therefore underlie the elaboration of geometry and all other disciplines deserving the title of "science." Again, only after ancient geometers had settled on the admissible constructions could they actually state their postulates. Likewise, before Kant can state any of *his* postulates, he must settle first on the constructive activities of the understanding. Thus one can imagine the Transcendental Analytic of the first *Critique* unfolding from 1772 to 1781 as follows.

First, Kant has to determine and distinguish from one another the activities that the understanding does in fact engage in. No doubt, the analysis of judgment, leading, as it did, to the derivation of the table of categories, can be understood to have served this purpose. But since it turns out, as a result of this analysis, that the fundamental activities of the understanding are merely "logical functions of judgment" which have "no meaning and [are] completely lacking in content" (A239/B298), a special argument has to be formulated in order to show that *these* activities are indeed constructive of anything at all. Nothing could be more surprising than to think, for example, that the activity

[26] I believe that these consideration also help answer a question put to me by Milos Arsenijevic. Why does Kant imagine that we have to have postulates at all? Does he not realize that Euclid's postulates might be reformulated as axioms? What do we have to gain or lose by such a reformulation?

of thought by which we distinguish subject and predicate—the activity that gives us the structure of every categorical judgment—could be *a priori* determining of any object. The ancient geometers do not have to be convinced that the activity of producing a line segment is constructive of line segments; they have only to be convinced that this constructive activity is simple or primitive enough that we may treat it as a starting point in geometry. So by contrast with the ancient geometers, Kant has to show that the activities of the understanding, however primitive they may be, are indeed constructive activites. We can take the project of a transcendental deduction to furnish just such an argument. Once that argument is given, and provided that it is successful, Kant can at once state and justify the postulates of empirical thought, which may then be put to use as one might put Euclid's postulates to use in geometry.[27]

Be *that* as it may, let me now make my second suggestion, which will bring me to the conclusion of my paper.

I argued earlier that, unless we conceive of reason, on Kant's view, as a faculty that sets problems—more especially the insoluble problem of finding systematic unity—scientific enquiry will be without either direction or motivation: scientific enquiry never stalls, because reason sets just those problems for us whose solution brings us closer and closer to systematic unity. But if we propose to solve problems set for us by reason, we must be permitted to act towards that end. This means more concretely that we must be allowed to implement a number of appropriately specified procedures. If, for example, the problem is to construct an equilateral triangle on a given line segment, we must be allowed to describe circles of any distance and center. Otherwise, it is all in vain. The function of a postulate is precisely to specify what we may do to the end of solving problems. When a problem has been set for us, we may then *rightly demand* that the appropriate protocol be implemented. Again, if the problem is to construct an equilateral triangle on any given line segment, I have every right to demand for my purpose the construction of circles of the required center and distance. For Euclid's third postulate permits me to construct circles of *any* center and distance. As Newton himself explains in the preface to the first edition of the *Principia*: "Geometry [unlike mechanics] does not teach us to draw [lines and circles], but *requires* them to be drawn, for it requires that the learner should first be taught [from mechanics] to describe these accurately before he enters upon geometry, *then it shows how by these operations problems may*

[27] This is, as Brigitte Falkenburg points out to me, going a little bit too fast. Kant cannot really justify his postulates of empirical thought, or any of the other principles of the understanding, before giving us the specific conditions under which they apply to a manifold of intuition. Thus the schematism will play a very important role in establishing the postulates of empirical thought as postulates in the strict sense and the other principles of the understanding as constructive procedures.

be solved. . . . It is the glory of geometry that from those principles [the postulates], brought [from mechanics], it is able to produce so many things [i.e., solve so many specifically geometrical problems]."[28] The significance of all this is that progress in geometry requires both problems and postulates. In a play on famous words by Kant, we can draw the moral of the story as follows. Postulates without problems have no point; problems without postulates are in vain. As a result, if Kant is to think of reason itself as a faculty of problems, he must have a separate faculty to give us the postulates and the constructive procedures licensed by the postulates needed for solving these problems. This separate faculty gives reason licence to do what it can to achieve its ends. It empowers reason,[29] but at the same time it keeps reason in line. Reason *cannot do just anything* to achieve its ends; it can only do rightly what it has been permitted to do by the understanding. If it acts within these bounds, it may expect to achieve results perhaps as brilliant as those of Archimedes.

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[28] Newton, *Principia*, trans. Motte and Cajori (Berkeley: University of California Press, 1962), xvii. Emphasis is mine.

[29] To the extent that reason is thus empowered by the postulates of the understanding, its tone of voice changes. If it sets a problem for us, its tone is interrogative. Once licensed by the postulates of empirical thought to take the appropriate measures to solve this problem, reason's tone becomes imperative. Kant's conception of reason and understanding as faculties of problems and postulates respectively explains this difference in tone of voice, which one notes from passage to passage in the *Critique of Pure Reason*.